

Exercise 6. (30pt)

(1) We have a signal $x(t)$.

(a) Express $x(t)$ in terms of the unit step function, then calculate $x'(t)$. (4 points)

(b) Plot the $x(-t + 2)$ and $x(\frac{2}{3}t + 1)$. (6 points)

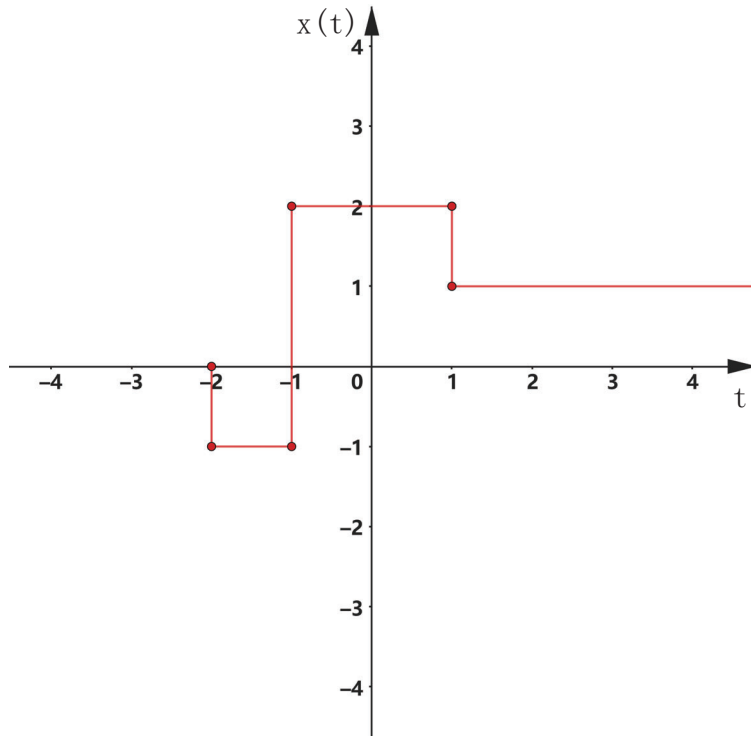


Figure 1: Signal $x(t)$ in P6(1)

(2) We have a signal $x(t)$, the following figures are the parts of $x(t)$ and its odd part $x_o(t)$, for $t \geq 0$ only. Please plot the whole odd part $x_o(t)$, whole even part $x_e(t)$ and whole $x(t)$ for $-\infty < t < \infty$ and write the equation of each function. (Be careful to write the boundary values clearly)

$$x(t) = \begin{cases} 1 + t, & 0 \leq t \leq 1, \\ 4 - 2t, & 1 < t \leq 2, \\ 0, & t > 2, \end{cases} \quad x_o(t) = \begin{cases} t, & 0 \leq t \leq 2, \\ -t + 4, & 2 < t \leq 3, \\ 1, & t > 3. \end{cases}$$

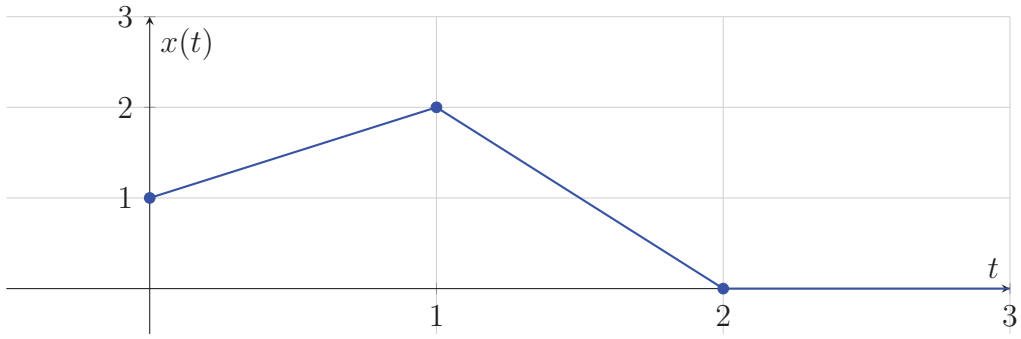


Figure 2: Original signal $x(t)$ for $t \geq 0$.

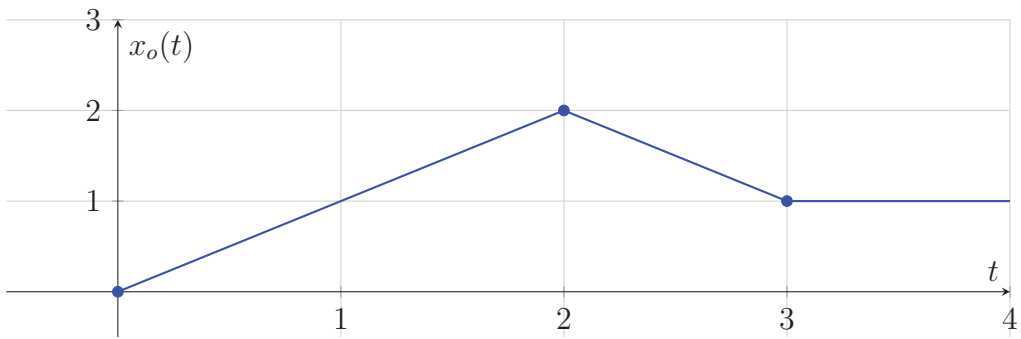


Figure 3: Odd part $x_o(t)$ for $t \geq 0$.

Solution: